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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.Tech (Construction Technology) (CBCS) Degree Examinations

May - June 2024

22CCT201 - CONSTRUCTION QUALITY AND SAFETY MANAGEMENT

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **One full** question from each **Unit**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Evaluate the effectiveness of quality circles in improving construction quality, considering factors such as employee engagement, problem-solving capabilities, and long-term quality outcomes.	10	L*3	1	1,6
b)	Construct a quality plan for a hypothetical construction project, incorporating elements such as quality objectives, performance metrics, and corrective action procedures.	10	L3	1	1,6
2. a)	Interpret the ISO 9000 family of standards in the context of construction quality management, emphasizing their role in establishing a systematic approach to quality assurance.	10	L3	1	1,2
b)	Formulate strategies for effectively implementing a quality system within a construction company, addressing challenges related to organizational culture, resource allocation, Continuous Monitoring and Improvement and stakeholder buy-in.	10	L3	1	1,3
Unit – II					
3. a)	Evaluate the impact of different approaches to managing time of completion in construction projects. How do efficient time management practices contribute to overall project quality?	10	L3	2	1,2
b)	Critically examine the role of codes and standards in quality planning for construction projects.	10	L3	2	1,4
4. a)	Analyze the inspection procedures commonly employed in the construction industry to ensure product and process quality.	10	L3	2	1,2
b)	Critically evaluate the components of a total quality assurance/quality control (QA/QC) program in the construction industry.	10	L3	2	1,2
Unit – III					
5. a)	Analyze the methods and techniques of quality assurance and quality control (QA/QC) in construction projects. How do these methods address the specific needs and challenges of ensuring quality throughout the project lifecycle?	10	L3	3	1,2,6
b)	Critically appraise the factors that influence construction quality, including organizational culture, workforce competence, and project management practices.	10	L3	3	1,6
6. a)	Critically assess any two quality improvement tools and techniques commonly used in construction, such as Six Sigma, Lean Construction, and Total Quality Management.	10	L4	3	3,6
b)	Evaluate the effectiveness of any two quality assurance objectives such as regularity agent, owner-oriented objectives, and design-oriented objectives in ensuring project quality. How do these objectives contribute to overall project success?	10	L3	3	1,3,6

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Unit – IV

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| 7. a) | Analyze the cost-effectiveness of implementing various safety measures on a construction site, including the use of personal protective equipment and training programs for workers. | 10 | L3 | 4 | 2,3 |
| b) | Evaluate the effectiveness of current safety policies and procedures in a construction company. Identify areas for improvement and propose specific strategies to enhance safety measures. | 10 | L3 | 4 | 2,4 |
| 8. a) | Apply the principles of safety management to a scenario where workers are exposed to lead poisoning during a construction project. Propose steps to eliminate or minimize the risk and ensure compliance with safety regulations. | 10 | L3 | 4 | 2,6 |
| b) | Evaluate the impact of lighting conditions on worker safety during excavation work. Determine optimal lighting requirements and recommend solutions to minimize accidents and improve visibility. | 10 | L3 | 4 | 1,2 |

Unit – V

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| 9. a) | How can safety surveys, inspections, sampling, and audits be effectively utilized to identify and mitigate potential safety hazards on a construction site? | 10 | L3 | 5 | 2,6 |
| b) | Examine the concept of "zero accidents" in construction projects. What are the key principles and challenges associated with implementing this concept? | 10 | L3 | 5 | 2,6 |
| 10. a) | Analyze the steps that a site engineer should take in response to an accident on a construction site. Consider immediate actions to ensure the safety of workers, documentation requirements, and communication protocols with relevant stakeholders. | 10 | L4 | 5 | 2,6 |
| b) | Evaluate the principles of accident prevention related to unsafe acts and conditions on construction sites. How can proactive measures such as hazard identification, safety training, and enforcement of safety protocols contribute to preventing accidents? | 10 | L4 | 5 | 2,6 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
